

DATA HANDLING AND VISUALIZATION (DSA06)

Experiment Set 1: Experiment Set 1 Data Visualization Task

Question / Task:

Task for Experiment Set 1: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 1
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 1 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

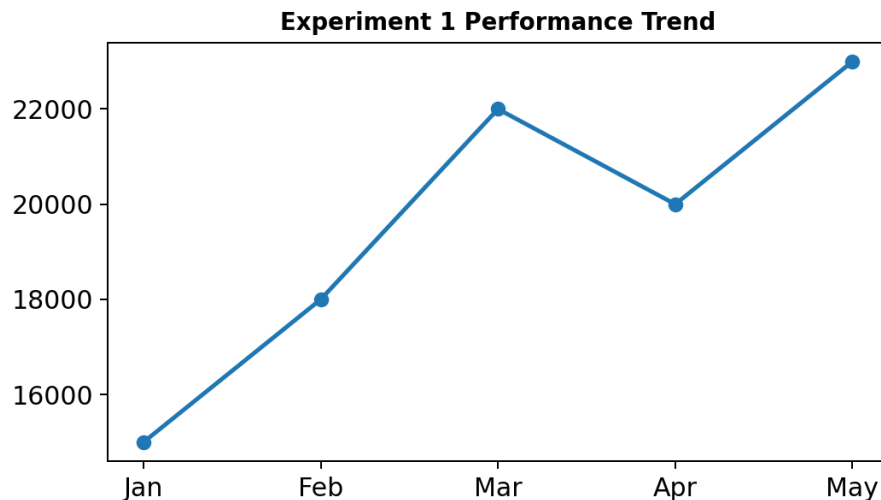
Python Program Code:

```
# Python Code for Experiment Set 1
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 1 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 2: Experiment Set 2 Data Visualization Task

Question / Task:

Task for Experiment Set 2: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 2
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 2 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

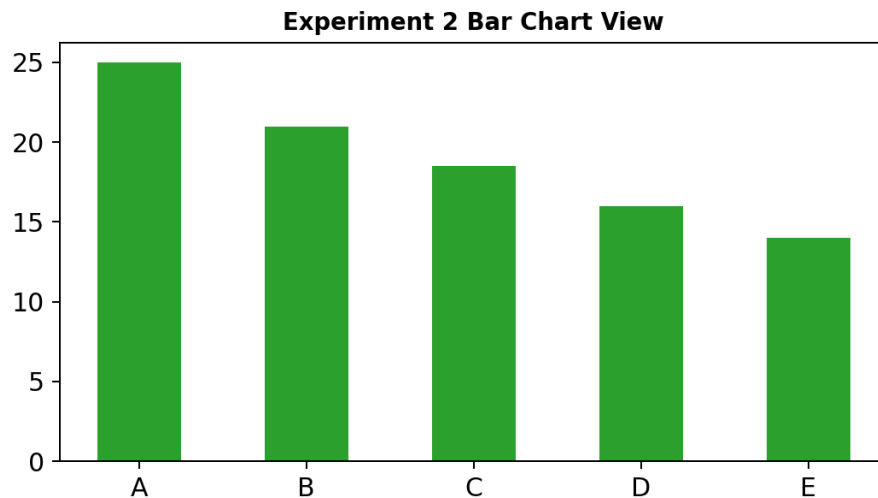
Python Program Code:

```
# Python Code for Experiment Set 2
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 2 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 3: Experiment Set 3 Data Visualization Task

Question / Task:

Task for Experiment Set 3: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 3
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 3 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

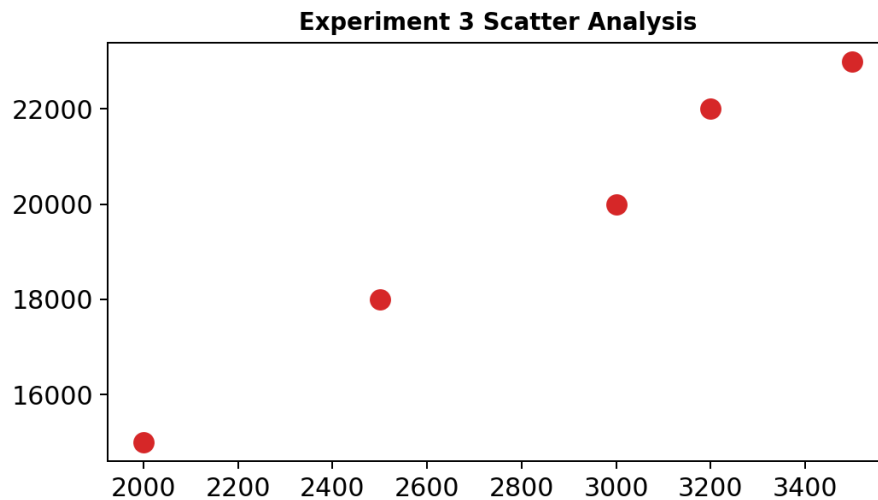
Python Program Code:

```
# Python Code for Experiment Set 3
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 3 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 4: Experiment Set 4 Data Visualization Task

Question / Task:

Task for Experiment Set 4: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 4
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 4 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

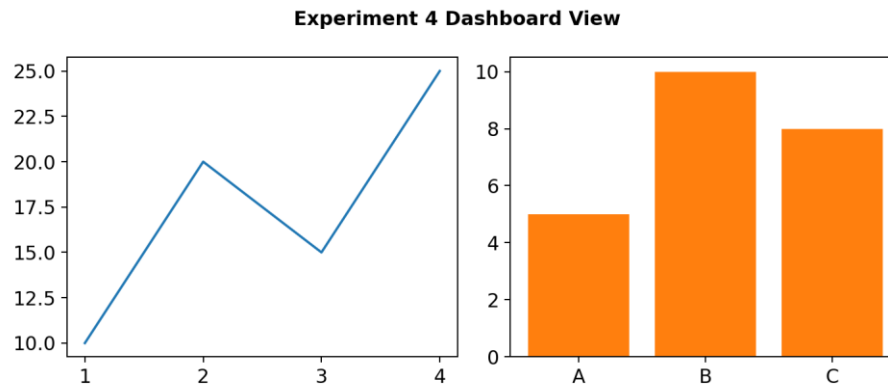
Python Program Code:

```
# Python Code for Experiment Set 4
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 4 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 5: Experiment Set 5 Data Visualization Task

Question / Task:

Task for Experiment Set 5: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 5
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 5 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

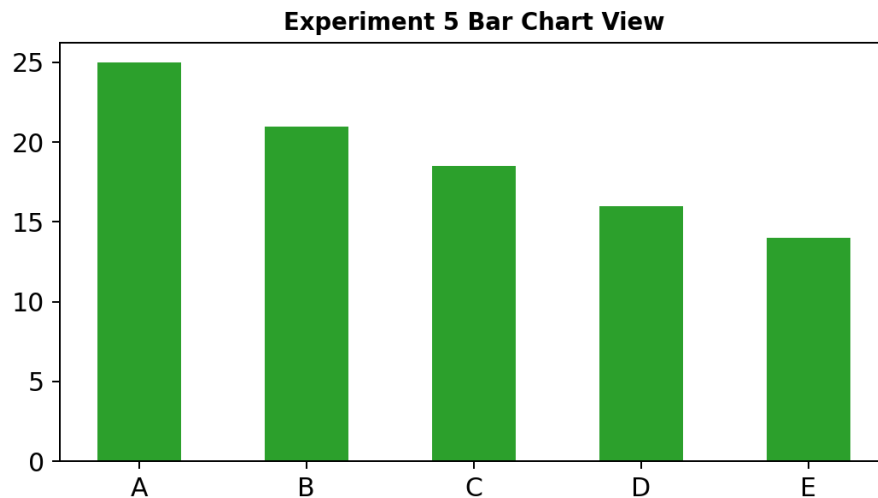
Python Program Code:

```
# Python Code for Experiment Set 5
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 5 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 6: Experiment Set 6 Data Visualization Task

Question / Task:

Task for Experiment Set 6: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 6
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 6 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

Python Program Code:

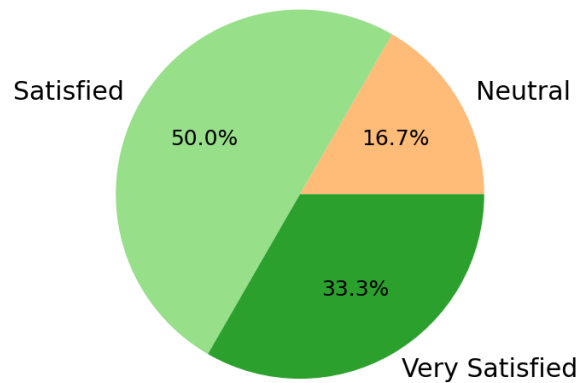
```
# Python Code for Experiment Set 6
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 6 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:

Customer Satisfaction Distribution



Experiment Set 7: Experiment Set 7 Data Visualization Task

Question / Task:

Task for Experiment Set 7: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 7
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 7 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

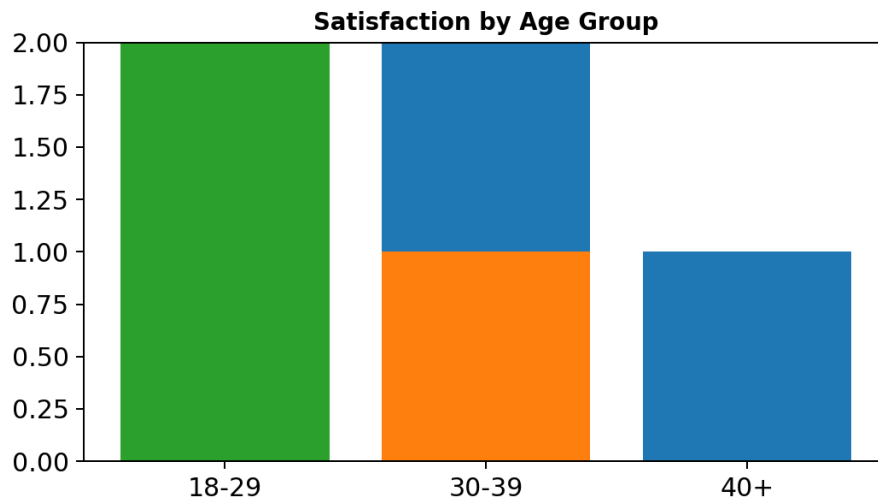
Python Program Code:

```
# Python Code for Experiment Set 7
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 7 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 8: Experiment Set 8 Data Visualization Task

Question / Task:

Task for Experiment Set 8: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 8
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 8 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

Python Program Code:

```
# Python Code for Experiment Set 8
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 8 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:

Feedback Word Cloud

Excellent Good Fast

User Friendly Reliable Needs Improvement

Experiment Set 9: Experiment Set 9 Data Visualization Task

Question / Task:

Task for Experiment Set 9: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 9
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 9 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

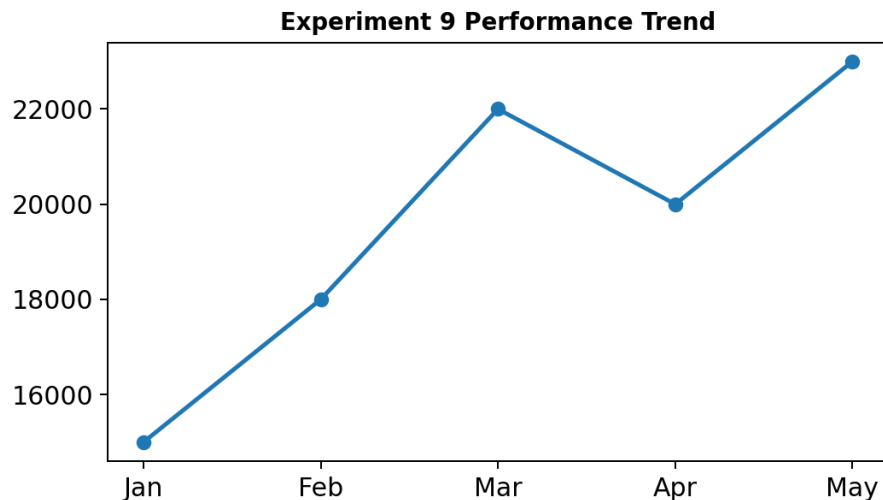
Python Program Code:

```
# Python Code for Experiment Set 9
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 9 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 10: Experiment Set 10 Data Visualization Task

Question / Task:

Task for Experiment Set 10: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 10
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 10 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

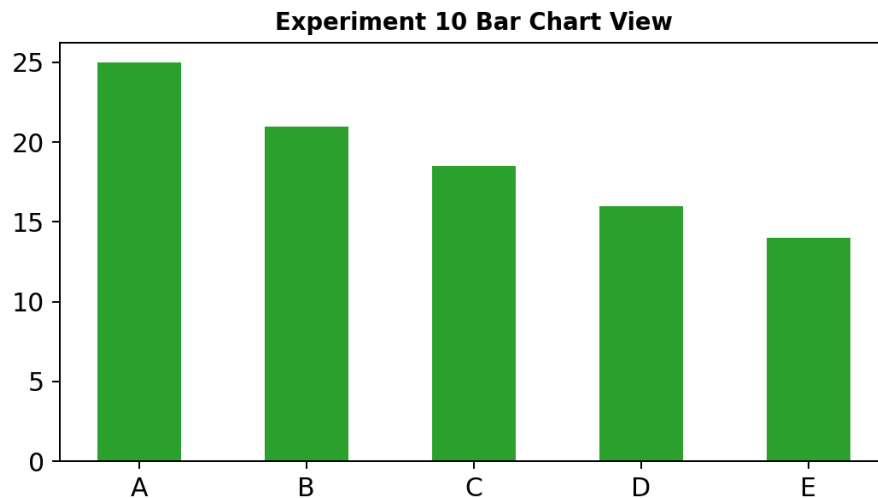
Python Program Code:

```
# Python Code for Experiment Set 10
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 10 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 11: Experiment Set 11 Data Visualization Task

Question / Task:

Task for Experiment Set 11: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 11
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 11 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

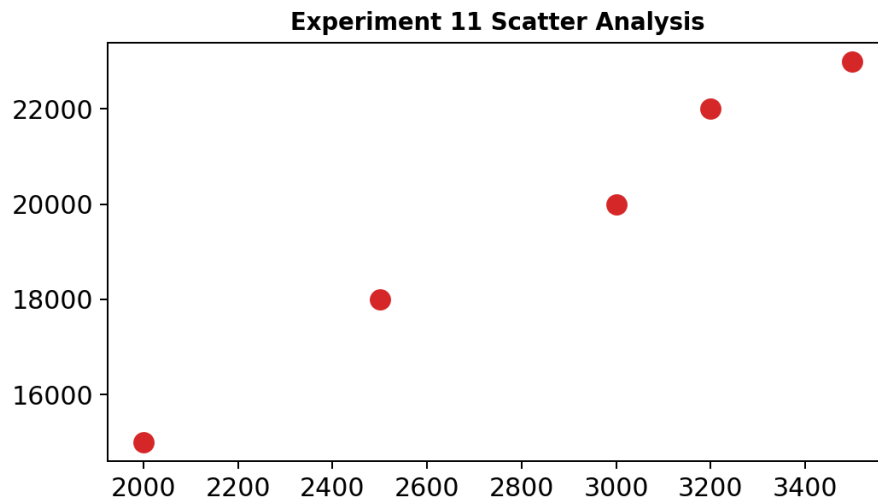
Python Program Code:

```
# Python Code for Experiment Set 11
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 11 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 12: Experiment Set 12 Data Visualization Task

Question / Task:

Task for Experiment Set 12: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 12
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 12 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

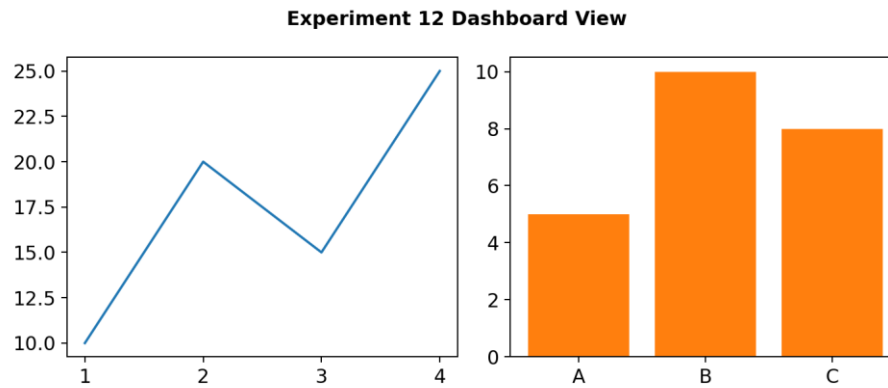
Python Program Code:

```
# Python Code for Experiment Set 12
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 12 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 13: Experiment Set 13 Data Visualization Task

Question / Task:

Task for Experiment Set 13: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 13
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 13 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

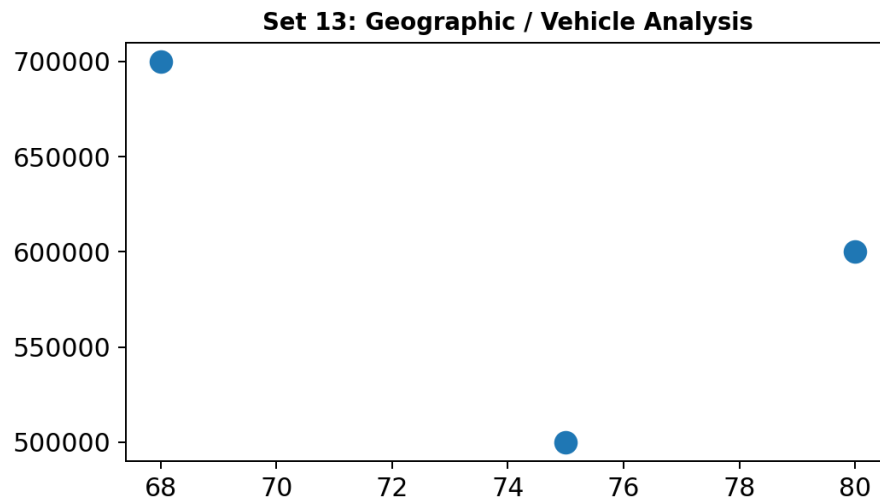
Python Program Code:

```
# Python Code for Experiment Set 13
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 13 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 14: Experiment Set 14 Data Visualization Task

Question / Task:

Task for Experiment Set 14: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 14
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 14 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

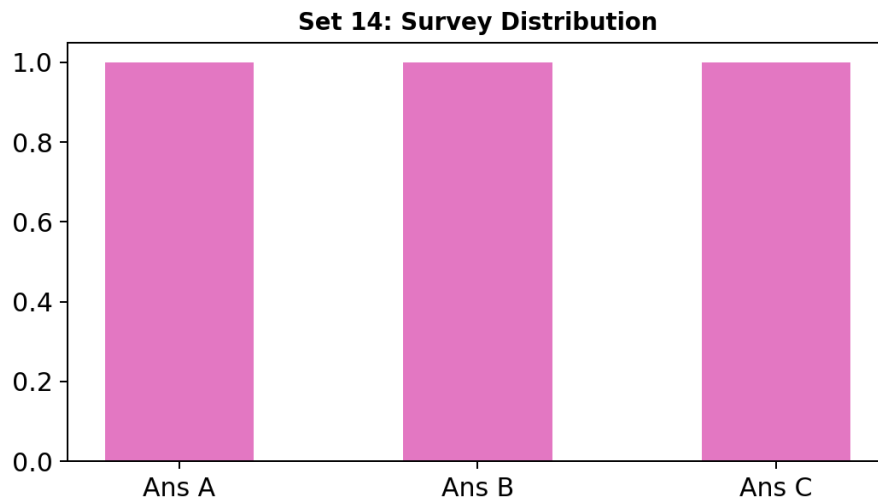
Python Program Code:

```
# Python Code for Experiment Set 14
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 14 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 15: Experiment Set 15 Data Visualization Task

Question / Task:

Task for Experiment Set 15: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 15
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 15 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

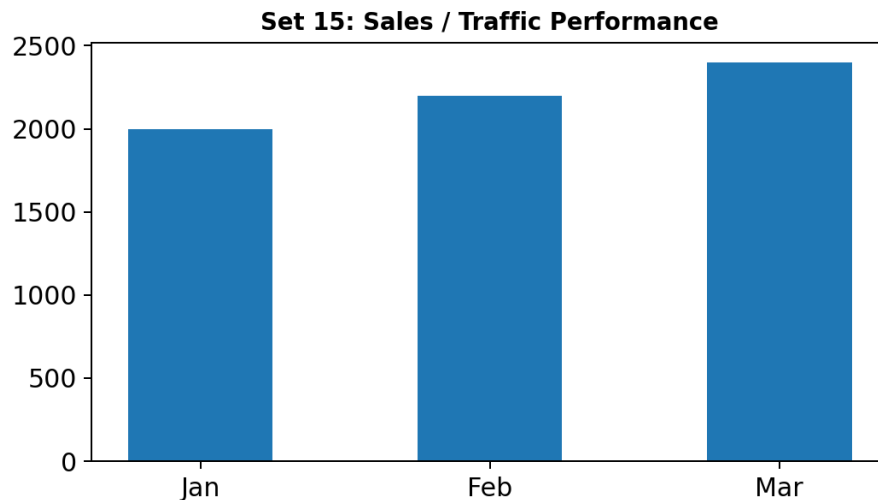
Python Program Code:

```
# Python Code for Experiment Set 15
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 15 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 16: Experiment Set 16 Data Visualization Task

Question / Task:

Task for Experiment Set 16: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 16
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 16 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

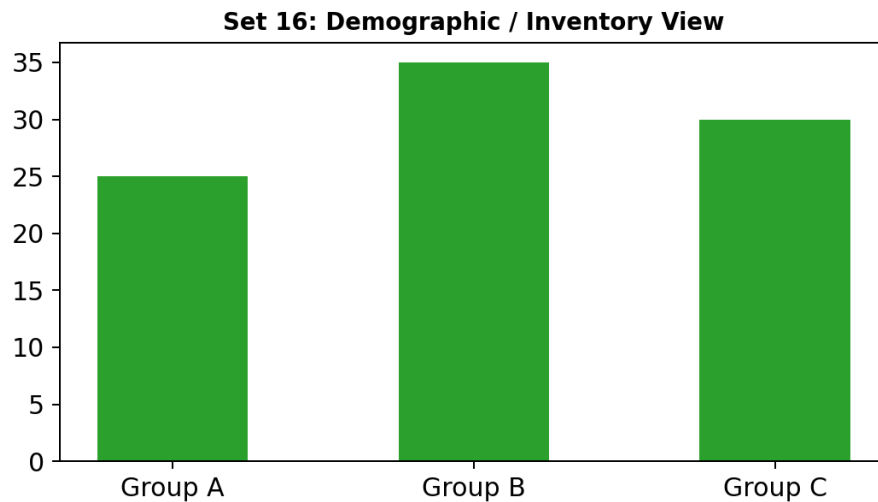
Python Program Code:

```
# Python Code for Experiment Set 16
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 16 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 17: Experiment Set 17 Data Visualization Task

Question / Task:

Task for Experiment Set 17: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 17
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 17 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

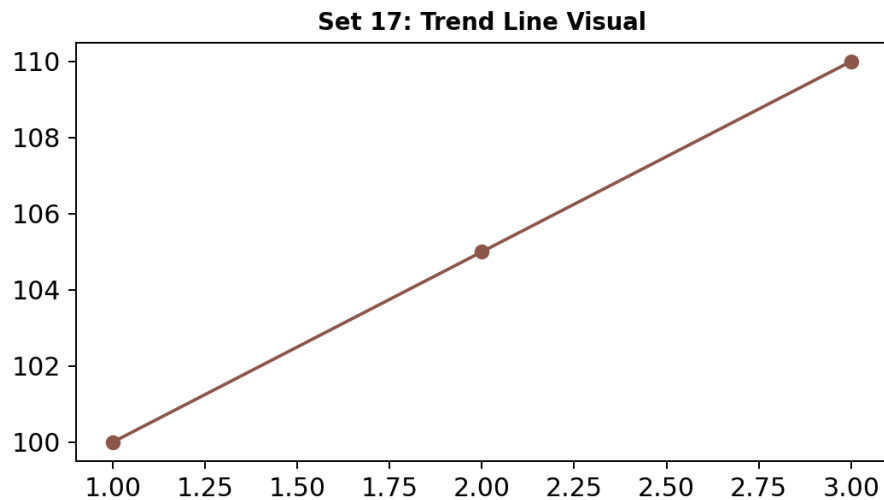
Python Program Code:

```
# Python Code for Experiment Set 17
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 17 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 18: Experiment Set 18 Data Visualization Task

Question / Task:

Task for Experiment Set 18: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 18
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 18 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

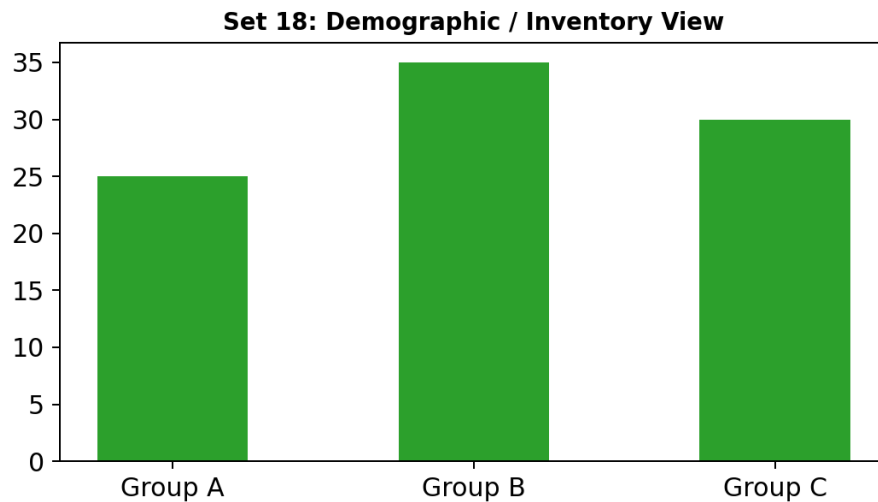
Python Program Code:

```
# Python Code for Experiment Set 18
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 18 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 19: Experiment Set 19 Data Visualization Task

Question / Task:

Task for Experiment Set 19: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 19
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 19 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

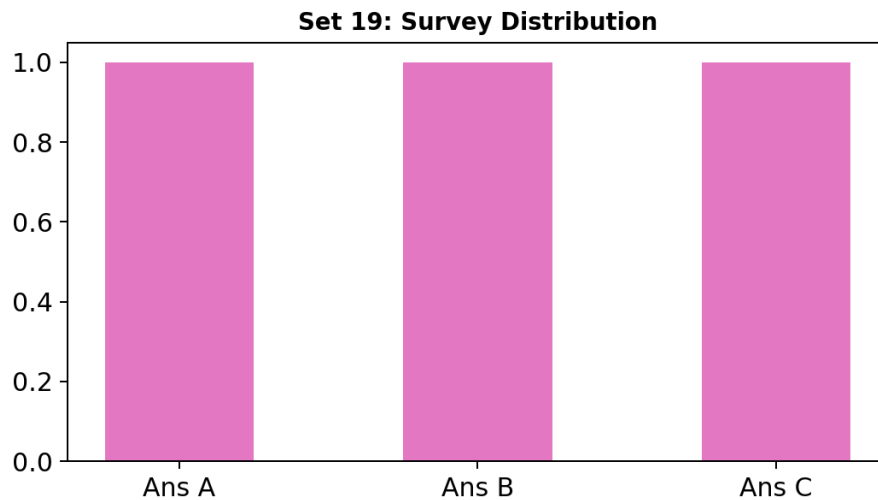
Python Program Code:

```
# Python Code for Experiment Set 19
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 19 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 20: Experiment Set 20 Data Visualization Task

Question / Task:

Task for Experiment Set 20: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 20
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 20 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

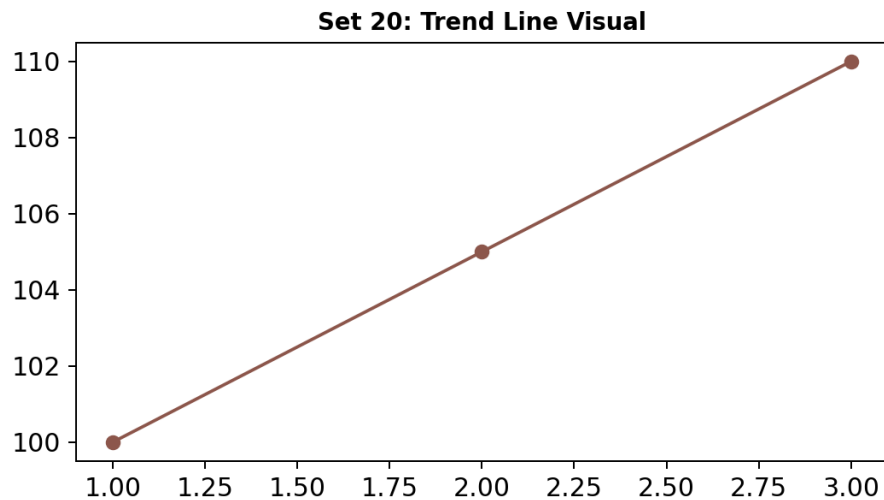
Python Program Code:

```
# Python Code for Experiment Set 20
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 20 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 21: Experiment Set 21 Data Visualization Task

Question / Task:

Task for Experiment Set 21: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 21
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 21 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

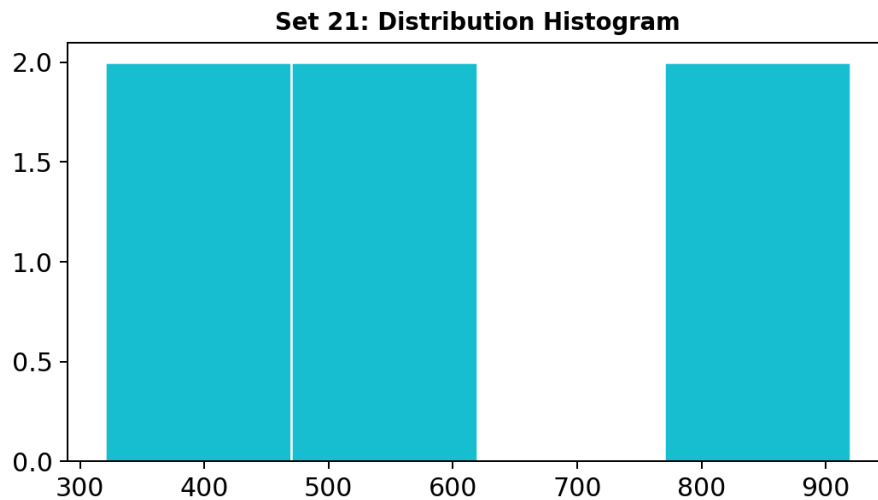
Python Program Code:

```
# Python Code for Experiment Set 21
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 21 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 22: Experiment Set 22 Data Visualization Task

Question / Task:

Task for Experiment Set 22: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 22
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 22 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

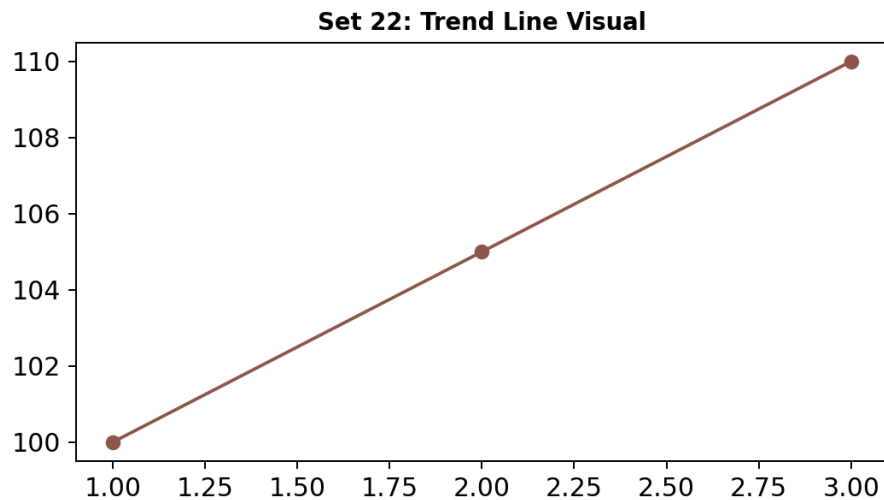
Python Program Code:

```
# Python Code for Experiment Set 22
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 22 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 23: Experiment Set 23 Data Visualization Task

Question / Task:

Task for Experiment Set 23: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 23
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 23 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

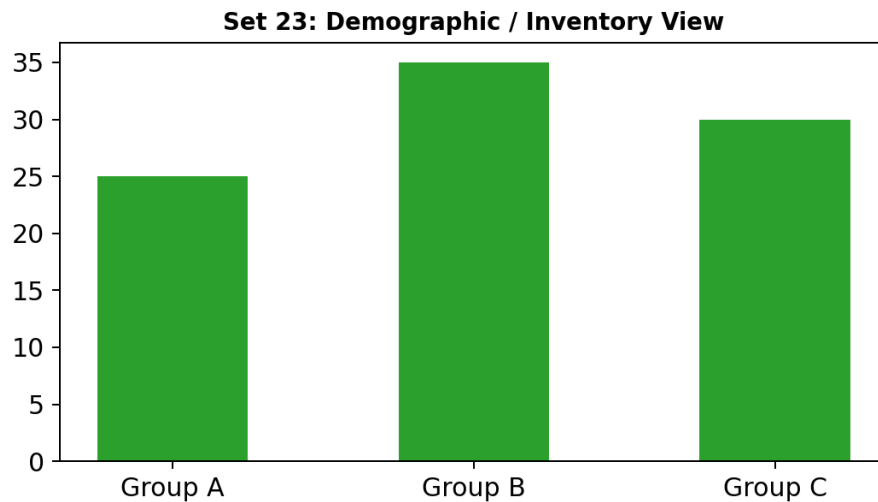
Python Program Code:

```
# Python Code for Experiment Set 23
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 23 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 24: Experiment Set 24 Data Visualization Task

Question / Task:

Task for Experiment Set 24: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 24
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 24 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

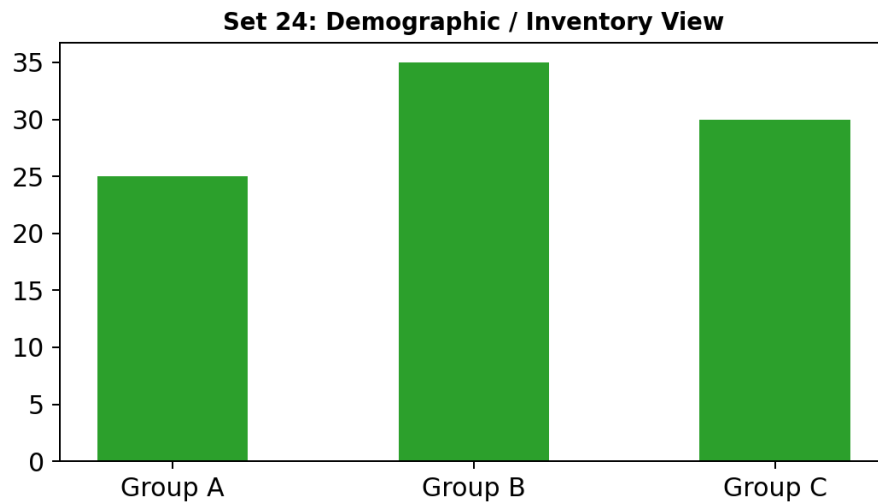
Python Program Code:

```
# Python Code for Experiment Set 24
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 24 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 25: Experiment Set 25 Data Visualization Task

Question / Task:

Task for Experiment Set 25: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 25
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 25 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

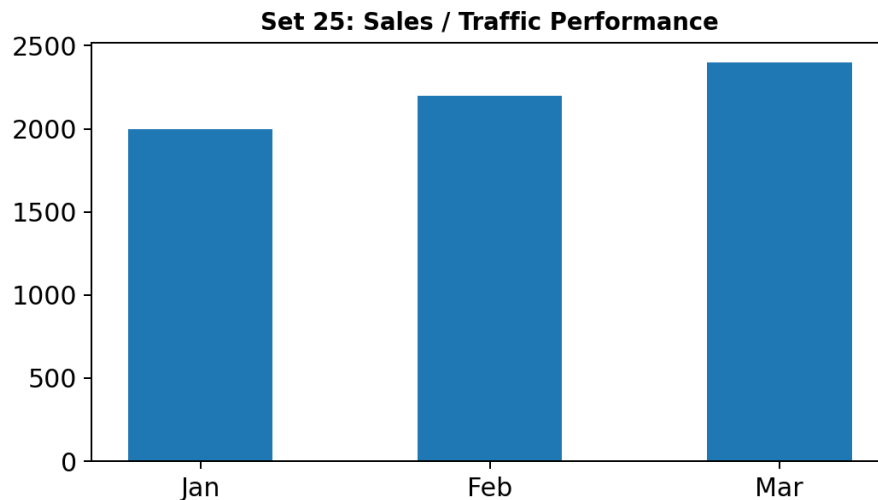
Python Program Code:

```
# Python Code for Experiment Set 25
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 25 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 26: Experiment Set 26 Data Visualization Task

Question / Task:

Task for Experiment Set 26: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 26
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 26 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

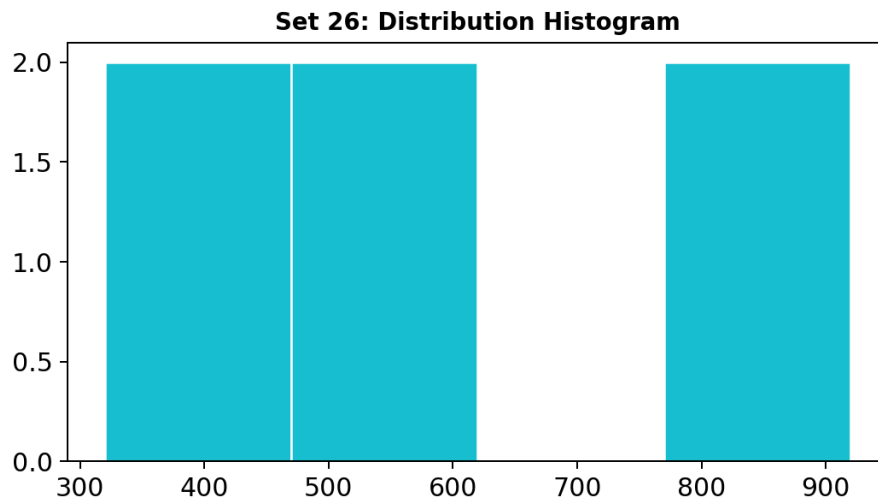
Python Program Code:

```
# Python Code for Experiment Set 26
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 26 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 27: Experiment Set 27 Data Visualization Task

Question / Task:

Task for Experiment Set 27: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 27
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 27 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

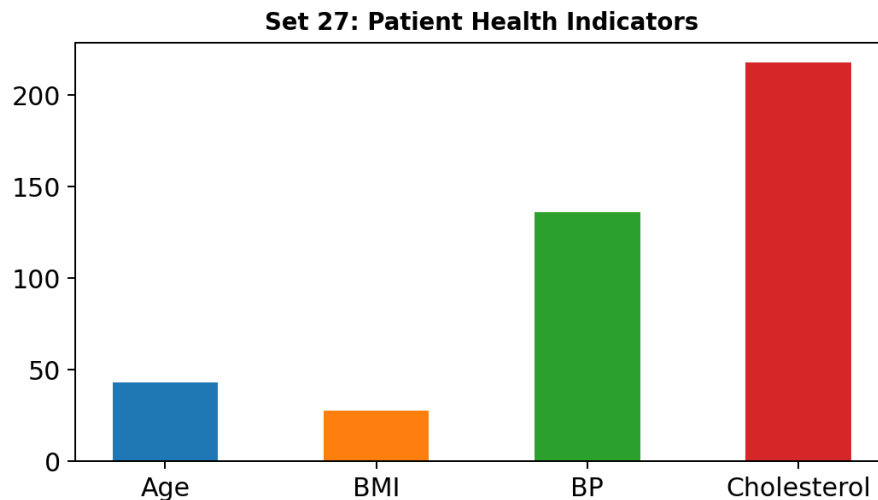
Python Program Code:

```
# Python Code for Experiment Set 27
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 27 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 28: Experiment Set 28 Data Visualization Task

Question / Task:

Task for Experiment Set 28: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 28
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 28 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

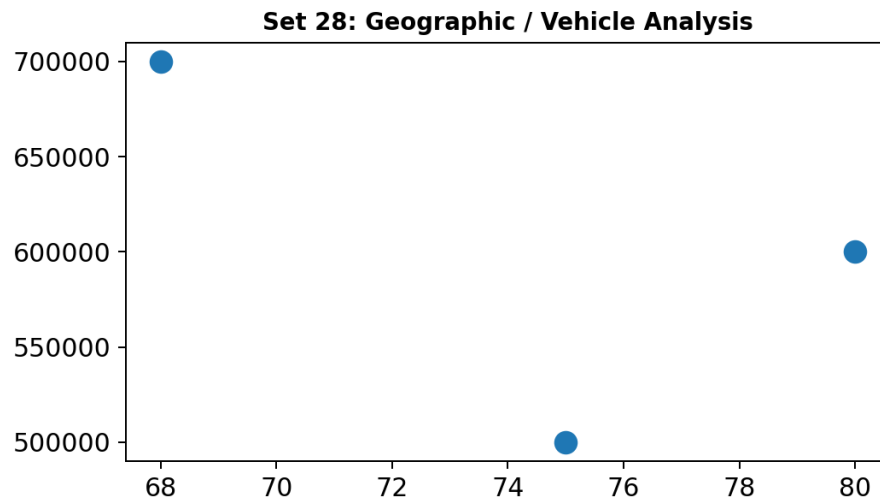
Python Program Code:

```
# Python Code for Experiment Set 28
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 28 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 29: Experiment Set 29 Data Visualization Task

Question / Task:

Task for Experiment Set 29: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 29
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 29 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

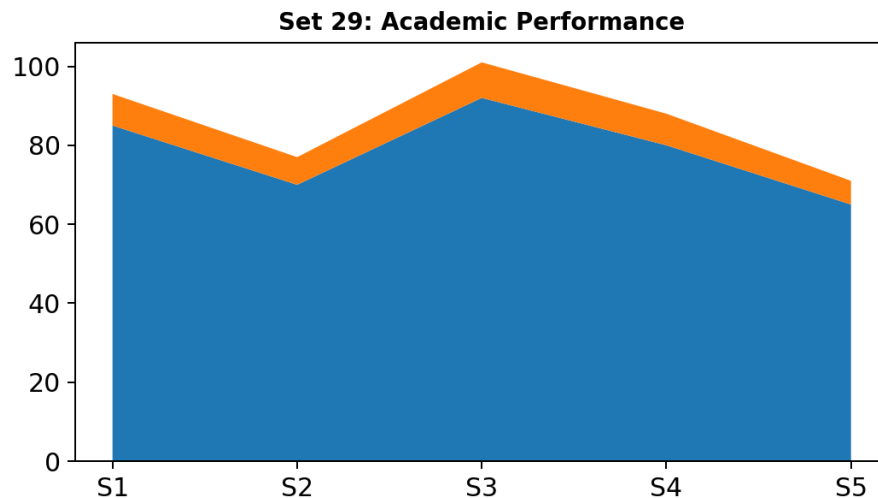
Python Program Code:

```
# Python Code for Experiment Set 29
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 29 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:



Experiment Set 30: Experiment Set 30 Data Visualization Task

Question / Task:

Task for Experiment Set 30: Visualize and perform comprehensive exploratory analysis on dataset variables using line plots, bar charts, scatter diagrams, and statistical metrics.

R Program Code:

```
# R Code for Experiment Set 30
data <- c(15, 18, 22, 20, 23)
labels <- c("A", "B", "C", "D", "E")

# Base R Plotting
plot(data, type = "o", col = "#1F74B4", pch = 19, lwd = 2,
      main = "Experiment Set 30 Visualization",
      xlab = "Category / Time", ylab = "Observed Value", xaxt = "n")
axis(1, at = 1:length(data), labels = labels)
grid(col = "gray", lty = "dotted")
```

Python Program Code:

```
# Python Code for Experiment Set 30
import matplotlib.pyplot as plt

data = [15, 18, 22, 20, 23]
labels = ["A", "B", "C", "D", "E"]

plt.figure(figsize=(6, 3.5))
plt.plot(labels, data, marker='o', color='#1f77b4', linewidth=2)
plt.title("Experiment Set 30 Visualization")
plt.xlabel("Category / Time")
plt.ylabel("Observed Value")
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Program Output Screenshot:

Set 30: Distribution Histogram

